

Research Paper

FCS-Net: Feather condition scoring of broilers based on dense feature fusion of RGB and thermal infrared images

Xiaomin Zhang^{a,b}, Tao Xu^{a,b}, Yanning Zhang^{a,b}, Yuan Gao^{a,b}, Jinming Pan^{a,b},
Xiuqin Rao^{a,b,*}

^a College of Biosystems Engineering and Food Science, Zhejiang University, Hangzhou, 310058, Zhejiang, China

^b Key Laboratory of Intelligent Equipment and Robotics for Agriculture of Zhejiang Province, Hangzhou, 310058, China

ARTICLE INFO

Keywords:

Poultry
Feather condition scoring
Image fusion
Infrared thermography
Deep learning

ABSTRACT

Assessing the feather condition of broilers is crucial for monitoring the animal welfare status and detecting the occurrence of feather pecking activities. Currently, the feather condition of individual broilers is manually scored by trained experts. To provide a more objective and efficient tool for feather condition scoring, a novel deep learning-based model, named Feather Condition Scoring Network (FCS-Net), was proposed based on RGB and thermal infrared images. The FCS-Net model combined the ResNet18 architecture with the proposed Dense Feature Fusion (DFF) module, which can effectively learn the feature mapping relationship between RGB and thermal infrared images. Before inputting the images into the network, an image registration process was conducted to align the RGB and thermal infrared images. The results showed that the FCS-Net model had a good performance for feather condition scoring, with the *Accuracy* of 97.02%, the *Precision* of 96.99%, the *Recall* of 97.04%, the F_1 of 97.01%, and the *Inference speed* of 15.34 fps. Compared to the ResNet18-RGB model, which only utilise RGB images, the FCS-Net model showed notable improvements in *Accuracy* by 4.02%, *Precision* by 3.90%, *Recall* by 4.08%, and F_1 by 4.01%. Moreover, it was observed that the FCS-Net model focused more on the back region of the broilers through heatmap visualization. Furthermore, the algorithm was compared with six typical image recognition algorithms including VGG16, ResNet18, SE-ResNet18, DenseNet121, Mobilenet_V2, and Shufflenet_V2.x1.0, as well as the state-of-the-art (SOTA) feather condition assessment methods. The results showed that the FCS-Net model achieved better performance than the six algorithms and the SOTA feather condition assessment methods. This study provided a valuable reference for automated monitoring of feather condition scoring of broilers in smart farming.

Nomenclature

Acronyms	
DFF	Dense Feature Fusion
FCS-Net	Feather Condition Scoring Network
FN	False Negative
FP	False Positive
FS	Feather scoring
Grad-CAM	Gradient-weighted Class Activation Mapping
IRT	Infrared thermography
ResNet18	18-layer ResNet Network
ResNet18_IR	IR-based ResNet18 Network
ResNet18_RGB	RGB-based ResNet18 Network
SGD	Stochastic Gradient Descent

(continued)

SOTA	state-of-the-art
TN	True Negative
TP	True Positive
Symbols	
C	Number of classes
F_1	Harmonic mean of precision and recall
Loss	Cross entropy loss
N	Number of samples
p_{ic}	Predicted probability of sample i belonging to class c
S_c	Output value of the last linear layer belonging to class c
$\sigma(S_c)$	Probability of the sample belonging to class c
y_{ic}	True value of sample i belonging to class c

(continued on next column)

* Corresponding author. College of Biosystems Engineering and Food Science, Zhejiang University, 866 Yuhangtang Road, Hangzhou City, 310058, Zhejiang Province, China.

E-mail address: xqrao@zju.edu.cn (X. Rao).

<https://doi.org/10.1016/j.biosystemseng.2024.09.002>

Received 21 September 2023; Received in revised form 23 August 2024; Accepted 2 September 2024

Available online 13 September 2024

1537-5110/© 2024 IAGrE. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

Feather cover condition serves as a significant indicator of the well-being of poultry (Xu et al., 2022). An intact feather cover ensures that chickens can move, fly, regulate their body temperature, and protect their skin effectively (Savory, 1995). However, feather loss in poultry can occur due to age, wear (Glatz, 2001), and feather pecking (Savory, 1995), typically affecting the rump and back regions (Bilcik & Keeling, 1999; Spindler et al., 2020). Poorly feathered chickens experience an elevated convective heat loss, resulting in increased heat production and higher feed consumption. Severe cases can lead to bleeding or infection in the damaged regions of the skin. (Lesson et al., 2004). These factors can potentially lead to economic losses for farmers (Decina et al., 2019). Therefore, feather condition scoring is essential for assessing the welfare status of poultry.

Currently, the most commonly utilised method for assessing feather condition is feather scoring (FS) (Tauson et al., 1984), which involves visually evaluating the feather cover on different parts of the body and assigning a subjective score. Researchers have proposed various feather scoring systems for assessing feather damage (Decina et al., 2019; Main et al., 2012; Butterworth, 2009). These scoring systems range from complete feather coverage to minor feather loss and substantial damage. The FS system provides an easy-to-use tool for assessing feather conditions. Nevertheless, this method is relatively subjective and less repeatable compared to objective assessment based on continuous variables such as radiated temperature (Cook et al., 2006).

Infrared thermography (IRT) is a non-invasive technique that detects and measures the heat radiation emitted from the surface of an organism (Cilulko et al., 2013). It provides a means to quantify and evaluate surface temperatures in the skin of animals, which are dependent on blood circulation (Redaelli et al., 2014). By observing increased circulation, IRT is potentially useful for inflammatory processes. This facilitates the identification of potential inflammatory processes or tissue damage based on variations in skin temperature (Casas-Alvarado et al., 2020). Feathers play a crucial role in providing insulation for poultry (Zhao et al., 2013). A broiler with inadequate feather coverage experiences more significant heat loss compared to one with good feather coverage (Mota-Rojas et al., 2021). Changes in skin temperature can be detected by IRT and determine whether the feathers are damaged. The areas of bare skin emit higher temperatures in the infrared spectrum compared to the surrounding feathered regions (McCafferty, 2013). Utilizing IRT to measure the surface temperature of chickens enables a more objective, accurate, and repeatable assessment of feather damage compared to the feather scoring method (Giersberg et al., 2017).

Recently, several studies have focused on assessing feather conditions in chickens using IRT. Cook et al. (2006) compared the effectiveness of FS and IRT-based methods for feather cover assessment in laying hens. The findings demonstrated that the IRT method provides a more objective and accurate assessment with continuous variables compared to the FS method. Similarly, Nääs et al. (2010) employed IRT to measure the surface temperatures of broilers at 42 days of age, revealing a significant correlation between the temperature radiated from the surface of the chickens and the feather coverage. They also observed that featherless areas were more sensitive to ambient temperature than feathered areas. Zhao et al. (2013) employed thermal infrared images to investigate the temperature distribution on the body surface of laying hens, enabling the identification of different feather conditions, such as excellent feather coverage, fair feather coverage, and no feather coverage, across various body parts of the hens. Pichová et al. (2017) employed IRT to evaluate feather damage in four different body parts of laying hens. They demonstrated that the areas with the most severe feather damage were the belly, back, and rump regions, which is consistent with previous research findings (Bilcik & Keeling, 1999). Additionally, the difference between the body surface and ambient temperature was positively correlated with the feather score, where a higher feather score indicated a more severe feather damage situation.

However, it is worth noting that the entire evaluation process in their study was conducted manually. Raine et al. (2020) observed that in areas above the threshold temperature of 33.5 °C (skin-exposed areas), there was a strong correlation between the feather score and the temperature difference (between the average body surface temperature and the ambient temperature of broilers). These findings demonstrated the potential of IRT as a valuable tool for assessing the feather condition of broilers.

In addition to detecting feather damage status based on single-source images, some scholars have also monitored feather conditions using multi-source fusion images. Deep learning or machine learning-based methods can provide a robust classification for feather condition assessment. Huang et al. (2020) presented a deep learning-based method of assessing feather conditions in laying hens using RGB and thermal infrared images. The model achieved an average accuracy of 83.9%, surpassing the accuracy obtained when using RGB images alone (82.4%) or thermal infrared images alone (79.0%). Lamping et al. (2022) proposed a feather-scoring network that fused visible images with depth images. The model achieved a mean average precision (mAP) of 98.02% for hen detection, and an accuracy of 91.83% for plumage condition scoring. Zhang et al. (2022) developed an approach to reconstruct RGB-Depth-Thermal maps using deep learning methods. The depth of feather damage can be assessed by the RGB-Depth-Thermal maps compared to the 2D RGB or thermal infrared images.

Existing studies have demonstrated that deep learning-based multi-source fusion methods can achieve good performance in feather condition assessment. Visible light offers observations that are closer to human visual perception, whereas thermal imaging can reveal potential inflammatory processes. Furthermore, visible light can provide clues to inflammation processes, such as redness. The fusion of these two modalities provides complementary information. However, current multi-source data fusion methods typically perform a simple fusion of visible light and thermal infrared information. For instance, the approach proposed by Lamping et al. (2022) fused the two images before the model input, while Huang et al. (2020) adopted a simple concatenation of the extracted feature vectors. These multi-source data fusion methods struggled to effectively learn the feature mapping relationships between different data sources, and therefore provided limitations in the performance of feather condition scoring and required enhanced generalization capabilities.

In this study, a novel model, named Feather Condition Scoring Network (FCS-Net) was proposed for feather condition scoring of broilers. The model was based on the residual block (He et al., 2016) and Dense Feature Fusion (DFF) of RGB and thermal infrared images. Before inputting the data into the FCS-Net model, image registration was performed to align the RGB and thermal infrared images, effectively addressing resolution and scale differences. By incorporating the DFF module, the feature mapping relationship between RGB and thermal infrared images can be learned, significantly enhancing the accuracy and generalization performance of the feather condition scoring method. Additionally, Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017) was employed to provide insights into the reasoning behind the classification rules in a more interpretable manner. This study can serve as a valuable reference for the intelligent feather condition scoring of broilers in commercial farms.

2. Materials and methods

2.1. Experimental design and data acquisition

The experiment was conducted in Huzhou City, Zhejiang Province, PR China. The animal study protocol was approved by the Laboratory Animal Welfare and Ethics Committee of Zhejiang University (No. ZJU20220214). Two groups of Chinese local yellow-feathered broilers were selected for the experiment, consisting of eight 24-week-old layers and twenty 20-week-old layers. The selected chickens possessed intact

feather coverage, normal body size, and foot tags as markers. Typically, these hens were raised free-range on a thickly bedded floor. During the data collection phase, they were housed in an individual cage, measuring 1.45 m long $\times$ 0.97 m wide $\times$ 1.30 m high. The cage was equipped with a drinker, a feed trough, and a manure collection trough (Fig. 1).

A dual-spectral thermography camera (Hikvision, DS-2TD2636-10, 8–14 μ m, uncooled core, China) was suspended at a height of 1.50 m above the floor of the cage. The dual-spectral thermography camera consisted of an RGB camera (6 mm, 1920 $\times$ 1080 pixels) and a thermal infrared camera (10 mm, 384 $\times$ 288 pixels). The distance between the RGB camera and the thermal camera was 42 mm. The dual-spectral thermography camera was connected to a network video recorder (Hikvision, DS-8608N-I8, China) for recording videos.

A total of 4 batches of video data were collected from yellow-feathered chickens on January 8, 2022, June 25, 2022, and July 3, 2022, respectively. These batches were labeled as B1, B2, and B3, as shown in Table 1. Batch B1 consisted of video clips from eight chickens, while B2 and B3 consisted of video clips from ten chickens respectively. Each chicken was recorded in multiple continuous videos within a 10-min timeframe between 12:00 and 17:00 p.m. The frame rate for all videos was set to 4 fps.

To simulate the feather damage of the chickens, the feathers of 18 chickens, including 8 chickens from batch B1 and 5 chickens from each of batches B2 and B3, were cut on the back area. The video data from batches B1 to B3 simultaneously acquired RGB and thermal infrared images. Thirty continuous video sequences were selected from multiple continuous videos recorded within a 10-min timeframe of a broiler. Each sequence had a duration of 4 s and a sampling frequency of 1 fps, resulting in 4 frames per sequence. This selection process yielded a total of 840 continuous video sequences from the 28 broilers, totaling 3360 frames.

2.2. Feather condition score and dataset

The feather condition score system utilised in this study referred to the assessment protocol (Butterworth, 2009), which involves a 3-point scoring system that encompasses the head, neck, back, rump, and belly regions. As only the back area was visible in RGB and thermal infrared images from the top view, scores from the back area were utilised for method development in this study. The detailed description of the feather scoring method is shown in Table 2 and the schematic diagram of different scores is shown in Fig. 2. When the yellow-feathered chickens have intact feather coverage without any noticeable feather damage, the feather condition is assessed as ‘Score 0’. When there are feather-damaged areas on the back of the yellow-feathered chickens, the damaged area is compared to a 5 cm diameter piece of black hard cardboard. If the diameter of the damaged area is less than 5 cm, the feather condition is assessed as ‘Score 1’. Otherwise, it is classified as

Table 1

The samples of broilers for feather condition scoring.

Date	Batch	Number of broilers	Week	Weight (kg)	Ambient temperature (°C)
2022/01/08	B1	8	24	1.20–1.65	7–16
2022/06/25	B2	10	19	1.05–1.65	25–29
2022/07/03	B3	10	20	1.10–1.65	28–35

Table 2

The detailed description of the feather scoring method.

Score level	Body conditions
0	Intact feather cover, no or slight wear, only single feathers missing
1	Damaged feathers (worn/deformed) or bald patch visible smaller than 5 cm in diameter
2	At least one bald patch visible greater than 5 cm in diameter

‘Score 2’.

The dataset of feather condition scoring is listed in Table 3, which includes 28 yellow-feathered chickens, of which 18 chickens were feather-damaged on their backs. A total of 840 pairs of continuous video sequences were used in the dataset, with 300 pairs classified as ‘Score 0’, 270 pairs as ‘Score 1’, and 270 pairs as ‘Score 2’. Each pair consisted of 4 set of binocular visible light images and 4 thermal infrared images. All images were manually labeled according to the assessment method described in Table 3. The 840 sequences were randomly divided into training, validation, and testing sets in a ratio of 7:1:2. Therefore, the training, validation, and testing sets comprised 588, 84, and 168 video sequences, respectively, corresponding to 2352, 336, and 672 frames. The dataset comprised diverse images of broilers, captured from various distances and angles, under different illuminations, and included images affected by chicken motion blur. Such diversity of the dataset contributed to a strong generalization capability of the FSC-Net network. Fig. 3 shows some typical images in the dataset.

2.3. Methodology

The overall pipeline for assessing the feather condition of chickens is illustrated in Fig. 4. Initially, RGB and thermal infrared images were captured through image acquisition. Given the significant differences in image resolution and scale between RGB and thermal infrared images, it was necessary to perform image registration to align these two types of images. Subsequently, the dataset was divided and utilised as input for the FCS-Net model to assess the feather condition. The ultimate result was the feather score of a chicken, reflecting the overall evaluation of the feather condition.

2.3.1. Image preprocessing

Before feeding the RGB and thermal images into the model, the thermal images were adjusted to achieve alignment with the RGB images. The image registration method utilised in this study was based on our published work (Zhang et al., 2022). This method relied on checkerboard images and effectively aligned RGB and thermal infrared images within a certain range of distances. The whole process of image registration consisted of two main steps: (i) scale and position adjustment and (ii) projection transformation. To expedite the classification of high-resolution images, a bilinear interpolation algorithm was employed to adjust the RGB image to a size of 960 $\times$ 540 pixels. The illustration of the thermal-RGB image registration is shown in Fig. 5. Through image registration, the thermal image (Fig. 5a) was adjusted to obtain the transformed thermal image (Fig. 5b). To demonstrate the effect of thermal-RGB image registration, a basic weighted fusion was

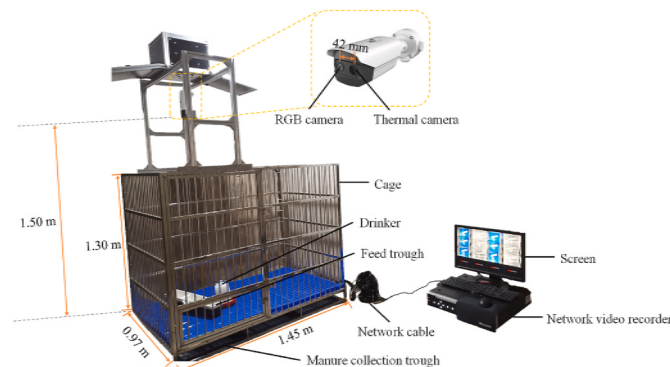


Fig. 1. Experimental setup and image acquisition.

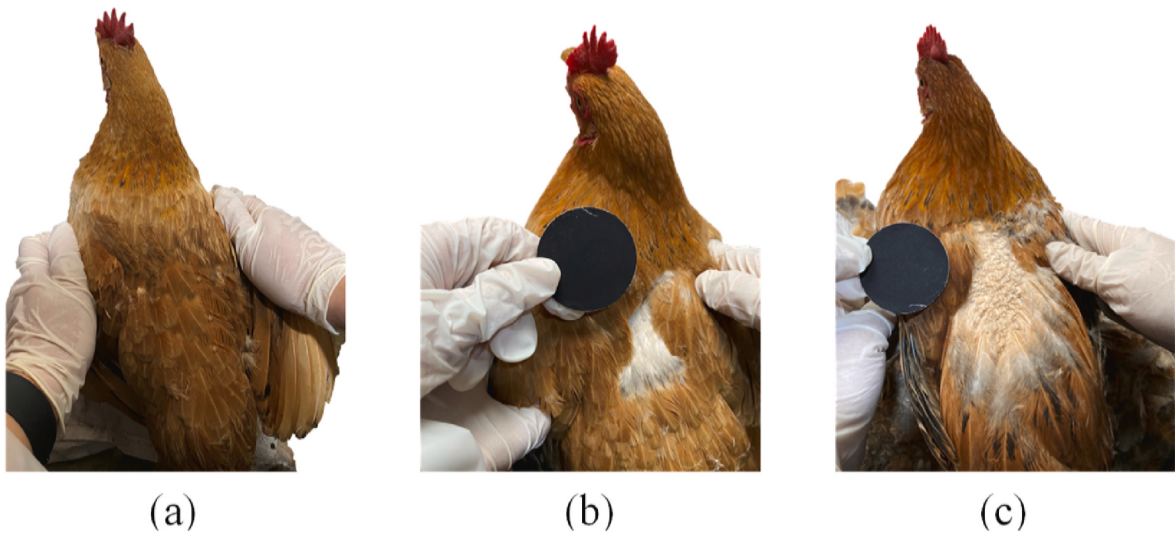


Fig. 2. Schematic diagram of different scores for feather condition scoring of broilers. (a) Score 0; (b) Score 1; (c) Score 2.

Table 3
Dataset of different feather condition scores for the broilers.

Score level	Number of broilers	Training set	Validation set	Testing set
0	10	840	120	240
1	9	756	108	216
2	9	756	108	216

performed to the RGB image (Fig. 5c) and transformed thermal image (Fig. 5b). It was observed that the majority of pixels in fusion image (Fig. 5d) was overlapped, indicating a good registration effect. The fusion of the RGB image and transformed thermal image was represented in Fig. 5d where the majority of pixels overlapped. Additionally, to facilitate subsequent feature extraction operation of the proposed network, all RGB and thermal infrared images were resized to 960×960 pixels.

2.3.2. Proposed model

The framework of the proposed Feather Condition Scoring Network (FCS-Net) is shown in Fig. 6. The upper and lower parts are the RGB and the thermal branches, respectively. The RGB branch and the thermal branch in the model had the same network architecture. The model was primarily composed of three components: the feature extractor, feature fusion module, and feature classifier.

The feature extractors for RGB and thermal infrared images adopt an 18-layer ResNet Network (ResNet18). ResNet18 introduces residual learning and incorporates residual blocks based on residual connections in the model architecture, effectively mitigating the accuracy degradation issue that arises with increased model depth. The proposed network utilised a dual-branch feature extraction structure based on ResNet18, taking RGB and thermal infrared images as input. Within the ResNet18 structure, four residual blocks (Res1, Res2, Res3, and Res4 modules) (Fig. 6) are connected sequentially. A convolutional layer with a stride of 2 is utilised in the final layer of each residual block to achieve feature map down-sampling and channel expansion. For each RGB or thermal infrared image, four different resolutions of feature maps with sizes of

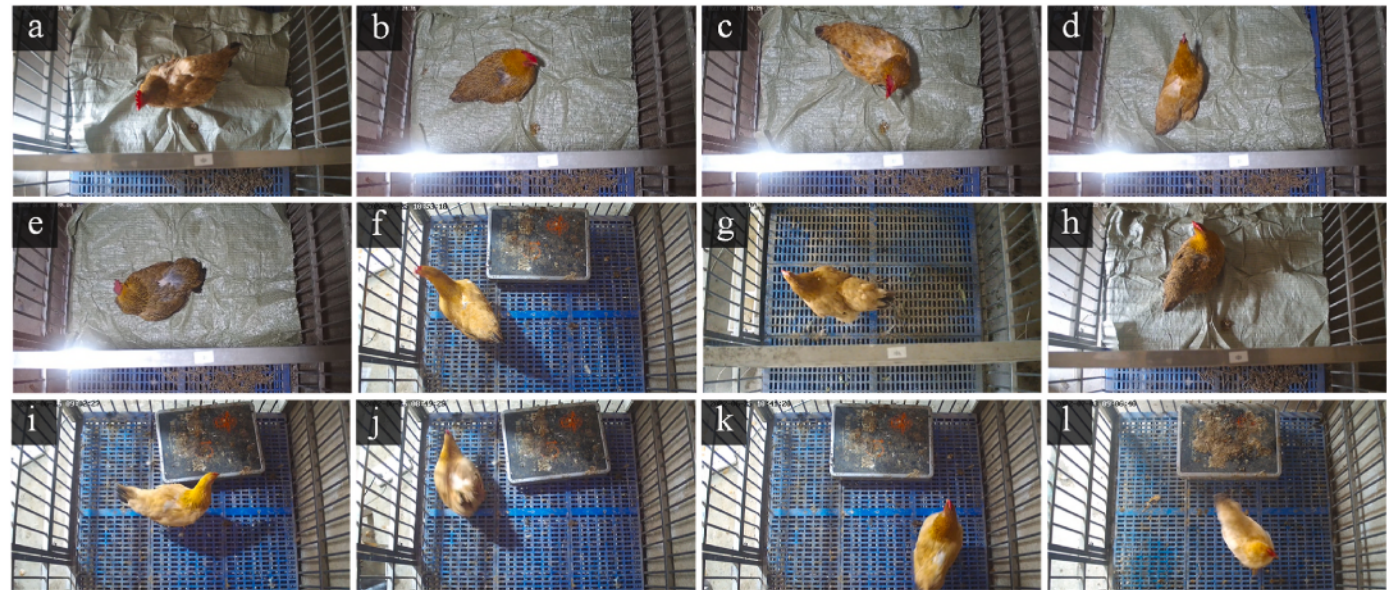


Fig. 3. Typical images in the chicken feather condition scoring dataset collected (a)–(d) from different angles, (e)–(h) under different illumination, and (i)–(l) with motion blur.

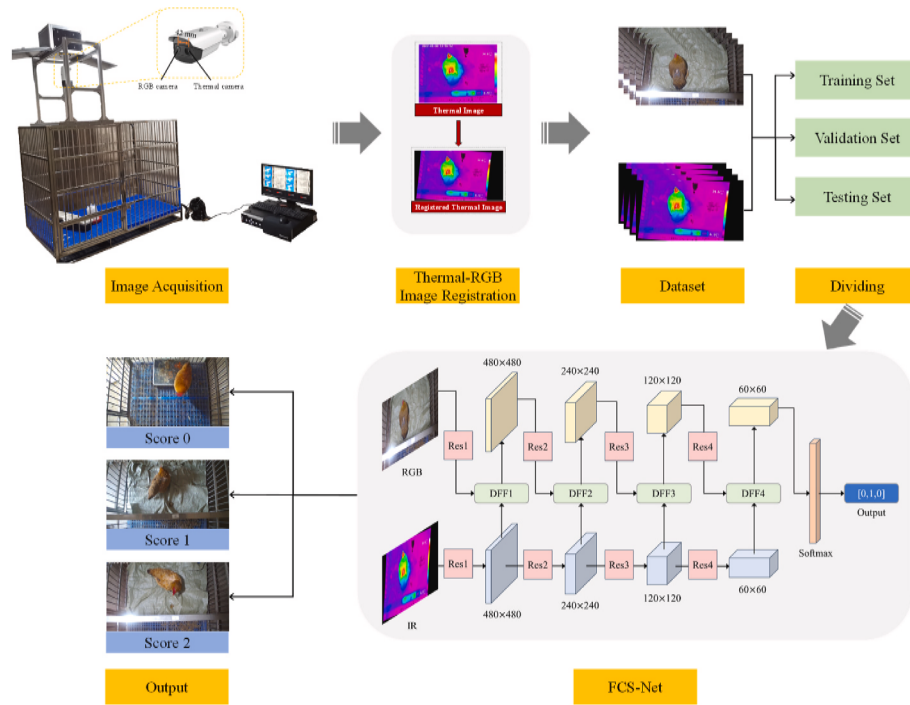


Fig. 4. The overview of the proposed Feather Condition Scoring (FCS-Net) model.

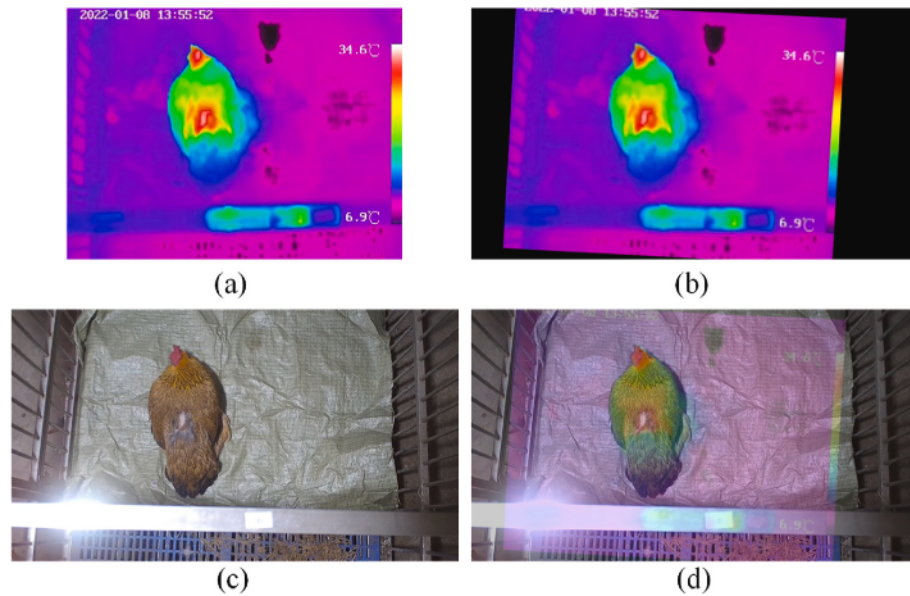


Fig. 5. The process of the image registration between RGB and thermal infrared image. (a) Original thermal infrared image; (b) Thermal infrared image after image registration from image (a); (c) Original RGB image; (d) Red-cyan anaglyph of images (b) and (c). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

480×480 pixels, 240×240 pixels, 120×120 pixels, and 60×60 pixels were generated from the feature extractor, respectively. Different levels of feature information were contained in these four feature maps with decreasing resolution and increasing channel dimensions.

The feature fusion module is the most crucial component of this network. Generally, feature fusion can be achieved through two approaches: (i) vector concatenation-based feature fusion (Fig. 7a) and (ii) multi-source same-scale dense feature fusion (Fig. 7b). The former simply concatenated the feature vectors without ensuring the consistency of the feature vector space between the two feature extractors. This lack of consistency may result in a decrease in the generalization

performance. In this study, Dense Feature Fusion (DFF) module was introduced between the RGB and the thermal branches to fuse different-scale dense features (Fig. 7b). Specifically, for each feature map output by a residual block in the feature extractor, the infrared feature map I_{IR} with dimensions of $W \times H \times C$ was first passed through an average pooling layer to obtain a feature vector m with dimensions of $1 \times C$. The feature vector m was then scaled by two fully connected layers to produce an output feature vector n with the same dimensions. Next, the feature vector n was multiplied element-wise with the RGB feature map I_{RGB} with dimensions of $W \times H \times C$, resulting in the fused output feature map I_{RGB_IR} , which combines visible light and thermal infrared

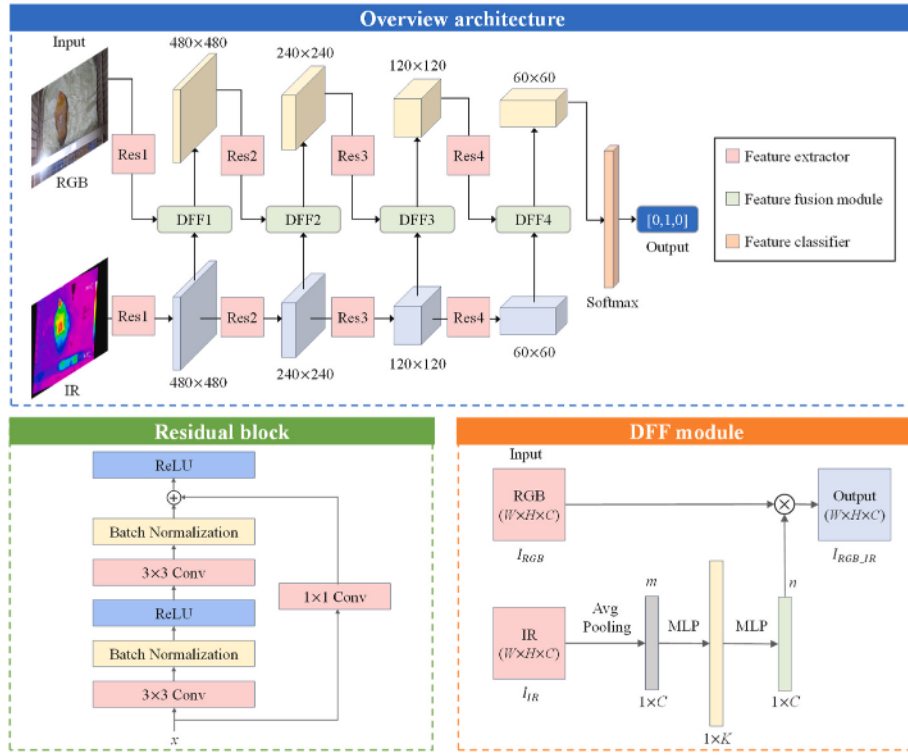


Fig. 6. Overview of the proposed Feather Condition Scoring (FCS-Net) model. The upper and lower parts represent the RGB and thermal branches, respectively. The RGB and thermal branches share identical network architecture configurations. The model consists of three main components: feature extractor, feature fusion module, and feature classifier. The feature extractors for RGB and thermal infrared images utilise an 18-layer ResNet Network (ResNet18). Four different resolutions of feature maps are generated from the feature extractor for each RGB or thermal infrared image. A Dense Feature Fusion (DFF) module is introduced between the RGB and thermal branches to effectively fuse dense features of different scales. The feature classifier section employs a Softmax classifier for classification purposes.

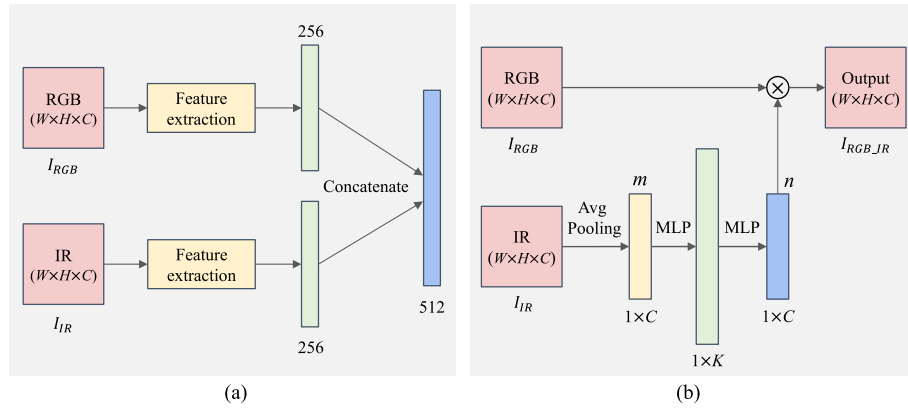


Fig. 7. Feature fusion method by (a) vector concatenating and (b) feature fusion method of multi-source homogeneous scale dense feature.

information. By performing operations in the feature fusion layer, the thermal feature map was superimposed with the RGB feature map, achieving the fusion of information from different modalities.

The feature classifier section employed the Softmax classifier (Joulin et al., 2017), which can normalise the S_c to a range between 0 and 1. The formula for the Softmax classifier is as follows:

$$\sigma(S_c) = \frac{e^{S_c}}{\sum_{j=1}^C e^{S_j}} \text{ for } c = 1, \dots, C \text{ and } S = (S_1, \dots, S_C) \in \mathbb{R}^C \quad (1)$$

Where $\sigma(S_c)$ represents the probability of the sample belonging to a certain class, S_c can take positive or negative values, and C denotes the number of classes.

The loss function used in this model was the cross-entropy function

(De Boer et al., 2005), which involves multi-class classification. The formula for the loss function is as follows:

$$\text{Loss} = -\frac{1}{N} \sum_i \sum_{c=1}^C y_{ic} \log(p_{ic}) \quad (2)$$

Where N is the total number of samples in the dataset, C is the number of classes, y_{ic} is the indicator function, such that $y_{ic} = 1$ when the true class of sample i is c , and $y_{ic} = 0$ otherwise, p_{ic} is the predicted probability of sample i belonging to class c .

2.3.3. Hyperparameters and experimental settings

To evaluate the effectiveness of the proposed feather condition scoring model FCS-Net, this study additionally trained two models based

on the ResNet18 network architecture. These models are named RGB-based ResNet18 Network (ResNet18_RGB), which only utilised visible light information, and IR-based ResNet18 Network (ResNet18_IR), which relied on thermal infrared information for feather condition scoring. The experimental setup and training parameters for these two models were consistent with the FCS-Net model. To interpret the key focus regions of different rating models, the feature maps of the last convolutional layer from ResNet18_IR, ResNet18_RGB, and FCS-Net models were visualised utilizing Grad-CAM.

All models used were trained from scratch. Stochastic gradient descent (SGD) was used as the optimiser. The initial learning rate is 0.01, and the decay rate is 0.5 for every 20 epochs. The model was trained for 100 epochs with a batch size of 16, a momentum of 0.9, and a weight decay of $1e-5$.

All models were trained on a deep learning platform equipped with an Intel® Xeon(R) Gold 6147M CPU @ 2.50 GHz, 256 GB RAM, and an Nvidia TITAN RTX GPU. The testing set measurements of three models were conducted on a desktop computer with an AMD Ryzen 9 4900 H S CPU @3.00 GHz, 16 GB RAM, and an Nvidia GeForce RTX 2060GPU. The software environment utilised “Python 3.7” as the programming language, and “PyCharm-Professional-2020.2” as the integrated development environment for image processing.

2.4. Performance metrics

To evaluate the performance of the proposed FCS-Net model, five metrics, i.e., *Accuracy*, *Precision*, *Recall*, F_1 , *Inference speed* (frames per second), and confusion matrix, were adopted for a comprehensive evaluation of model performance. The calculation formulas of metrics are as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (6)$$

where *Accuracy* is the proportion of the correctly predicted feather scores to all predicted feather scores, *Precision* is the proportion of the correctly predicted positive feather scores to all predicted positive feather scores, *Recall* is the proportion of correctly predicted positive feather scores to all actual positive feather scores, F_1 is the harmonic mean of *Precision* and *Recall*, True Positive (TP) is the number of positive feather scores predicted as positive, True Negative (TN) is the number of negative feather scores predicted as negative, False Positive (FP) is the number of positive feather scores predicted as positive, and False

Negative (FN) is the number of positive feather scores predicted as negative.

3. Results

3.1. Model performance

The loss curves and accuracy curves of the ResNet18_RGB, ResNet18_IR, and FCS-Net models are shown in Fig. 8. Around 100 iterations, all three models had basically converged. Notably, the FCS-Net model demonstrated a faster convergence rate, and its training loss approached 0 more closely in comparison to the other two models. The FCS-Net model also achieved significantly higher *Accuracy* on the validation set, with ResNet18_RGB being the second-best, and ResNet18_IR showing the lowest accuracy. Furthermore, the accuracy curve of the FCS-Net model remained more stable in the later iterations, indicating superior classification performance and enhanced robustness.

The performance metrics of the three models on the testing set are shown in Table 4. It can be seen that the FCS-Net model performed the best on the testing set, followed by the ResNet18_RGB model, and the ResNet18_IR model showing the lowest performance. This may be due to the true labels for feather condition scores being obtained from visible light images, leading to a higher correlation between visible light images and the true labels. The results showed that the FCS-Net model had a good performance for feather condition scoring, with the *Accuracy* of 97.02%, the *Precision* of 96.99%, the *Recall* of 97.04%, the F_1 of 97.01%, and the *Inference speed* of 15.34 fps on the testing set. Compared to the ResNet18_RGB model, which performed relatively well among the other two models, the FCS-Net model improved *Accuracy* by 4.02%, *Precision* by 3.90%, *Recall* by 4.08%, and F_1 by 4.01%. These results indicated that the FCS-Net model had a higher classification accuracy and stronger robustness for feather condition scoring. Additionally, the fusion of visible light and thermal infrared images effectively enhanced the performance of the model.

3.2. Model confusion matrix

The confusion matrix of the ResNet18_RGB, ResNet18_IR, and FCS-Net models is illustrated in Fig. 9, providing insights into the classification performance for each class. It can be observed that both the

Table 4

Comparison of the *Accuracy*, *Precision*, *Recall*, F_1 , and *Inference speed* of three models for feather condition scoring task.

Model	Accuracy (%)	Precision (%)	Recall (%)	F_1 (%)	Inference speed (fps)
ResNet18_RGB	93.00	93.09	92.96	93.00	17.38
ResNet18_IR	87.95	88.20	87.90	87.99	17.56
FCS-Net	97.02	96.99	97.04	97.01	15.34

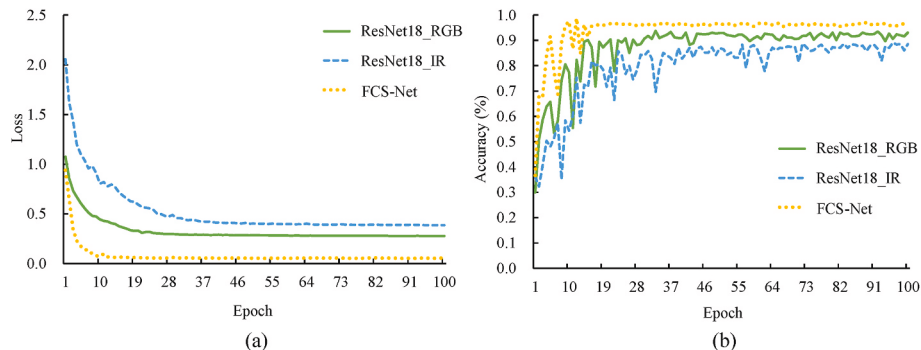


Fig. 8. Comparison of the results on training and validation sets. (a) Training loss function curve; (b) Accuracy curve on the validation set.

ResNet18_RGB and ResNet18_IR models tended to misclassify samples labeled as ‘Score 1’ as ‘Score 0’. This might be attributed to the small size of the damaged areas, posing challenges in detecting feather damage regions solely from RGB or thermal infrared images. Additionally, the ResNet18_RGB and ResNet18_IR models tended to misclassify samples labeled as ‘Score 2’ as ‘Score 1’. This may be related to the posture of broilers. For instance, when broilers are in ‘Spreading wings’ posture, the pixel area of damaged feathers is larger, whereas in ‘Standing’ and ‘Walking’ postures, the area of damaged feathers is smaller. Therefore, the posture of broilers is a significant factor leading to the misclassification of samples labeled as ‘Score 2’ as ‘Score 1’.

3.3. Interpretability of the feather condition score model

The class activation maps of the ResNet18_RGB, ResNet18_IR, and FCS-Net models are shown in Fig. 10, each score including an RGB image, a thermal infrared image, and three class activation maps generated by the three models. It can be observed that different models exhibited distinct class activation maps for the same score category, indicating varying key regions of focus. The ResNet18_IR model primarily focused on the left background area in thermal infrared images of all three score categories, whereas the ResNet18_RGB and FCS-Net models paid more attention to the broiler itself, which may elucidate why the performance of the ResNet18_IR model was slightly inferior to the other two models. The potential reason why the ResNet18_IR model paid more attention to the background region could be attributed to the fact that the last convolutional layer of the model already produced pseudo-coloured images. When visualizing with Grad-CAM, it was difficult to focus on the texture features of the broiler itself, instead paying more attention to the background region. Specifically, for ‘Score 0’, the ResNet18_RGB model highlighted the head and tail regions of broilers, while the FCS-Net model focused more on the head area of broilers. For ‘Score 1’ and ‘Score 2’, the ResNet18_RGB model emphasised the head or tail regions of broilers, whereas the FCS-Net model concentrated more on the back area of broilers. These findings were aligned with human cognition and understanding of feather-

damaged areas, contributing to the superior performance of the FCS-Net model.

3.4. Comparison of different image recognition algorithms

Deep learning-based classification algorithms have gained significant popularity among researchers for various object recognition tasks. To validate the effectiveness of the FCS-Net model, the proposed method was compared with six image classification algorithms, including VGG16, ResNet18, DenseNet121, SE-ResNet18, Mobilenet_V2, and Shufflenet_V2_x1_0, to evaluate the performance in feather condition scoring for the broilers (He et al., 2016; Hu et al., 2018; Huang et al., 2017; Ma et al., 2018; Sandler et al., 2018; Simonyan & Zisserman, 2014). To conduct the comparison, each of the aforementioned six algorithms was trained on the training set of RGB images separately. Subsequently, their performance on feather condition scoring was evaluated on the testing set. The *Accuracy*, *Inference speed*, and the number of parameters of seven models were compared in this experiment. The corresponding experimental results are illustrated in Table 5 and Fig. 11.

The results indicated that VGG16 had the lowest *Accuracy* (90.50%), the largest number of parameters (134.28 M), and the intermediate *inference speed* (13.31 fps). It consumed more computational resources without delivering significant improvements in accuracy or speed, rendering VGG16 suboptimal as a model choice. Densenet121 had a higher *Accuracy* (94.50%), but its *Inference speed* (10.45 fps) was the lowest. MobileNet_V2 and Shufflenet_V2_x1_0 had advantages in the number of parameters, with 3.23 M and 1.26 M, respectively. Their *Accuracy* and *Inference speed* were quite similar. Compared to ResNet18, SE-ResNet18 introduces a channel attention module (SE module) (Hu et al., 2018), which dynamically adjusts the importance of each channel, thereby enhanced the feature representation capabilities. Therefore, SE-ResNet18 achieved a higher *Accuracy* (94.00%) and a lower *Inference speed* (15.76 fps) compared with ResNet18. Overall, MobileNet_V2, Shufflenet_V2_x1_0, SE-ResNet18, and ResNet18 showed comparable performance, but there was still potential for improving the *Accuracy*. By

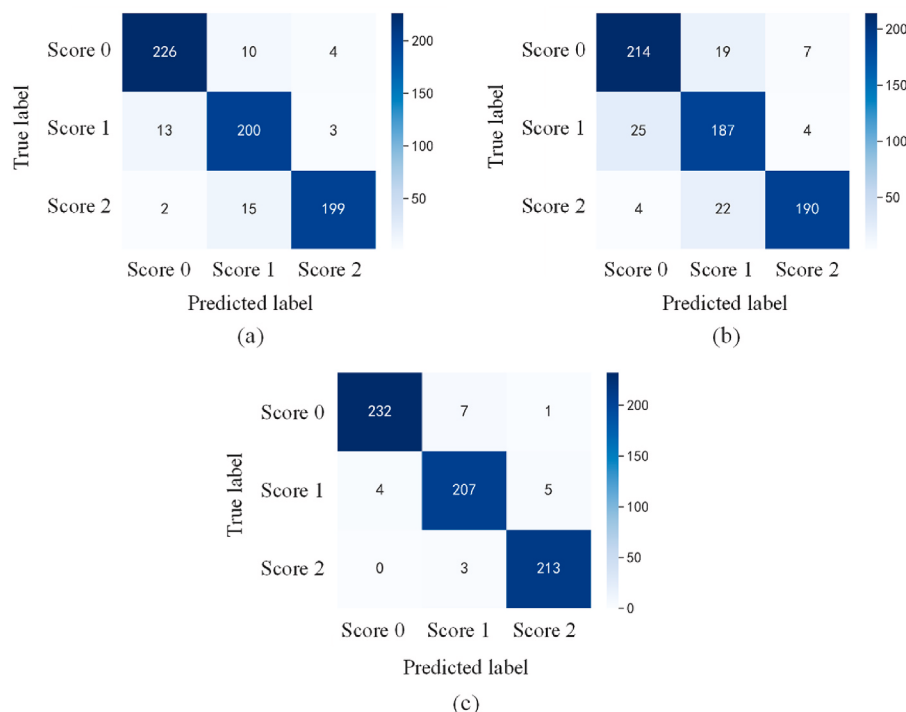


Fig. 9. Confusion matrix of three models for feather condition scoring in yellow-feathered chickens. (a) ResNet18_RGB model, (b) ResNet18_IR model, (c) FCS-Net model. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

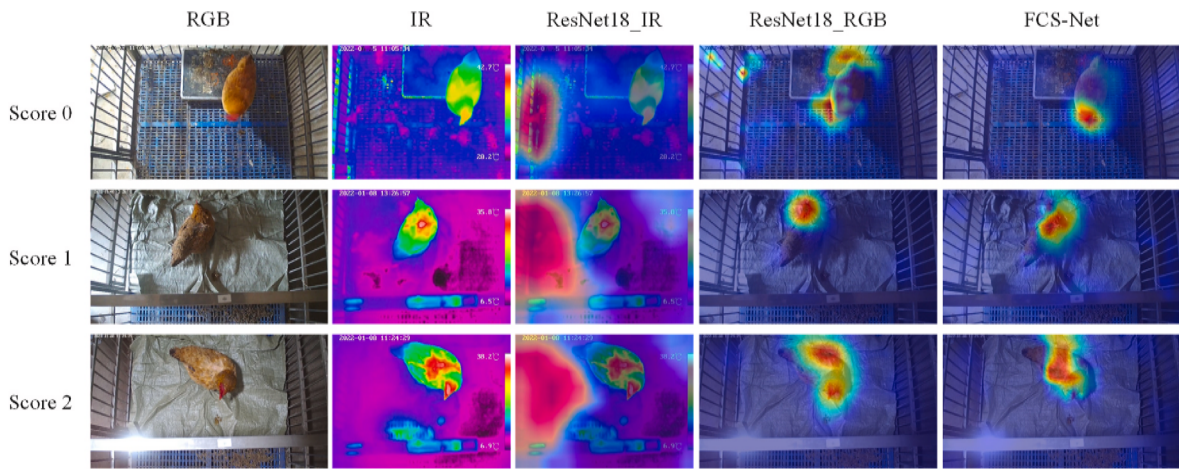


Fig. 10. Comparison of the class activation maps for ResNet18_RGB, ResNet18_IR, and FCS-Net models towards feather condition scoring of broilers.

Table 5

Comparison of the proposed algorithm with six excellent image recognition algorithms.

Model	Accuracy (%)	Inference speed (fps)	Parameters (M)
VGG16	90.50	13.31	134.28
ResNet18	93.00	17.38	11.18
SE-ResNet18	94.00	15.76	11.27
DenseNet121	94.50	10.45	7.86
Mobilenet_V2	93.45	15.22	3.23
Shufflenet_V2_x1_0	93.60	16.48	1.26
Ours	97.02	15.34	20.13

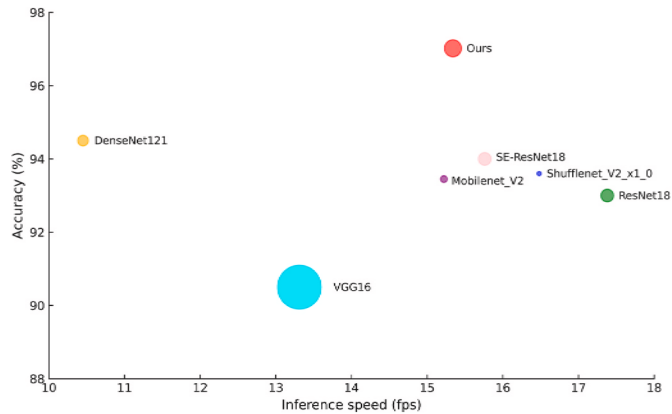


Fig. 11. The bubble plots of Accuracy, Inference speed, and the number of parameters of six image recognition models and the Feather Condition Scoring Network (FCS-Net) model (Ours).

fusing visible light and thermal infrared information through the DFF module, the proposed FCS-Net model showed the highest Accuracy (97.02%) in feather condition scoring for the broilers, with a significant improvement. It also maintained a good balance between Accuracy and Inference speed (15.34 fps). The number of parameters (20.13 M) remained acceptable. These results demonstrated that the model proposed in this study was of practical significance for feather condition assessment.

4. Discussion

4.1. Comparison with state-of-the-art (SOTA) feather condition assessment research

The research related to feather condition scoring based on multi-source data fusion and computer vision is shown in Table 6. Huang et al. (2020) were among the first to apply the idea of multi-source data fusion to the research of feather condition scoring for broilers. However, their approach of simple feature concatenation resulted in a less satisfactory Accuracy of 83.9%. Differently, Lamping et al. (2022) utilised RGB and depth images for feather condition scoring, adding an extra output layer after Mask R-CNN (He et al., 2017) to predict the feather scores. Nevertheless, the results indicated that incorporating depth information did not lead to a significant improvement in overall scoring performance. This may be attributed to the fact that depth information alone cannot be directly used as a determinant of feather damage. Conversely, thermal infrared images are effective in evaluating potential inflammation or tissue damage processes due to changes in skin temperature. Zhang et al. (2022) reconstructed a chicken model combining RGB, depth, and thermal infrared information from stereo colour and thermal infrared cameras. The model was capable of estimating the depth of feather damage with an R^2 of 0.946. However, the depth of damage was not integrated into the feather scoring system. In comparison to the aforementioned studies, the proposed FCS-Net model introduced the DFF module, effectively integrating the RGB branch and the thermal branches, overcoming the limitations of detecting feather damage characteristics using single-spectrum information. The experimental results demonstrated that integrating thermal infrared information significantly improved the Accuracy (97.02%) of feather condition scoring compared to the aforementioned studies. Furthermore, FCS-Net exhibited stronger robustness, overcoming challenges

Table 6

Comparison of the proposed algorithm with the state-of-the-art feather condition scoring research.

Author	Publication	Method	Inputs	Accuracy (%)	R^2
Huang et al.	ACPLF 2020	Xception	RGB + IR	83.9	–
Lamping et al.	COMPAG 2022	Mask-RCNN	RGB + Depth	91.83	–
Zhang et al.	Animals 2022	3D reconstruction	RGB + Depth + IR	–	0.946
Ours	–	ResNet18+DFF	RGB + IR	97.02	–

posed by complex backgrounds and varying lighting conditions in poultry farms.

4.2. Limitations and future works

The experimental results demonstrated that the FCS-Net model provided a powerful tool for feather condition scoring of broilers. The proposed method had significant advantages regarding the aspects of inference speed and lighting adaptability. In terms of inference speed, the proposed method achieved an inference speed of 15.34 fps, enabling real-time monitoring for feather conditions scoring of broilers. In terms of lighting adaptability, RGB cameras can utilise infrared night vision function for illumination at night, while thermal imagers are unaffected by changes in lighting conditions. Therefore, the proposed model in this study effectively can overcome variations in lighting conditions.

Limitations can be found in this paper. In the simulated flat-rearing conditions used in this study, each image only involved a single chicken. However, in practical commercial environments, group feeding is predominant. Compared to single-target scenarios, situations involving multiple broilers often encounter challenges such as frequent movements of chickens and mutual obstruction. These factors pose difficulties for feather condition scoring in multi-broiler scenarios. As a result, the model cannot effectively assess the feather coverage of chickens that are obstructed. Additionally, the posture of chickens can influence feather damage recognition, and the score of identified feather-damaged regions varies with different postures.

Therefore, future works will be focused on optimizing and transferring the model to handle occlusions and posture variations of multiple chickens. To adapt our approach for practical commercial environments, subsequent efforts can be involved augmenting the existing network architecture by adding a multi-target detector to the RGB branch's head for multiple broilers monitoring. The multi-target detector, such as YOLOv7 (Wang et al., 2023), can be fine-tuned to localise the positions of each chicken in images and recognise their postures. If the posture of the chicken is recognised as “standing”, it can be segmented from the RGB and thermal infrared images based on the localization results and used as input to FCS-Net for subsequent feather condition scoring. Additionally, datasets containing images of broilers with different feather colours and real-world feather injuries will be incorporated for model fine-tuning and iteration, thereby enhancing its generalizability across various feather colours and forms of feather injuries of broilers. Considering the computational requirements and system costs associated with deploying deep learning models in practical applications, the proposed model will be deployed on embedded devices, such as Jetson Nano, to achieve stable and efficient real-time monitoring for broilers. This research will be further pursued in subsequent studies to address the mentioned challenges and to ensure the adaptability of our system in commercial environments.

5. Conclusion

Feather condition scoring is essential for assessing the welfare status of poultry. In this study, a novel deep learning-based model, named FCS-Net, was proposed for feather condition scoring of broilers. The model utilised a combination of the ResNet18 architecture and the proposed DFF module, effectively learning the feature mapping relationship between RGB and thermal infrared images. Image registration was conducted to align the RGB and thermal infrared images. By incorporating the DFF module, FCS-Net had a good classification accuracy for feather condition scoring, with an *Accuracy* of 97.02%, a *Precision* of 96.99%, a *Recall* of 97.04%, a F_1 of 97.01%, and an *Inference speed* of 15.34 fps. Compared with the ResNet18_RGB model that only utilised RGB images, the FCS-Net model demonstrated improvements in *Accuracy* by 4.02%, *Precision* by 3.90%, *Recall* by 4.08%, and F_1 by 4.01%. Grad-CAM was employed to identify the key regions where the model focused. It can be noticed that the FCS-Net model placed greater emphasis on the back

region of the broiler. Furthermore, the algorithm was compared with six typical image recognition algorithms including VGG16, ResNet18, SE-ResNet18, DenseNet121, Mobilenet_V2, and Shufflenet_V2_x1_0, as well as the SOTA feather condition assessment methods. The results showed that the FCS-Net model achieved better performance than the six algorithms and the SOTA feather condition assessment methods. The proposed method had significant advantages regarding the aspects of inference speed and lighting adaptability. This study provided a valuable reference for the intelligent feather condition scoring of broilers. In future work, the FCS-Net model would be further optimised to suit the practical environment and commercial applications. Subsequent efforts can be involved optimizing and transferring the model to handle occlusions, posture, feather colour variations of multiple chickens. Additionally, the proposed model will be deployed on embedded devices to achieve stable and efficient real-time monitoring for broilers.

CRedit authorship contribution statement

Xiaomin Zhang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tao Xu:** Writing – review & editing, Validation, Formal analysis. **Yanning Zhang:** Visualization, Software, Data curation. **Yuan Gao:** Writing – review & editing, Investigation, Data curation. **Jinming Pan:** Supervision, Resources. **Xiuqin Rao:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Key R&D Program of Zhejiang Province (No. 2022C02024). The authors appreciate the funding organisation for their financial supports. The authors would also like to thank the helpful comments and suggestions provided by all the authors cited in this article and the anonymous reviewers.

References

- Bilcik, B., & Keeling, L. J. (1999). Changes in feather condition in relation to feather pecking and aggressive behaviour in laying hens. *British Poultry Science*, 40(4), 444–451. <https://doi.org/10.1080/00071669987188>
- Butterworth, A. (2009). *Welfare Quality® assessment protocol for Poultry (broilers, laying hens)*. the Netherlands: Welfare Quality® Consortium, Lelystad.
- Casas-Alvarado, A., Mota-Rojas, D., Hernández-Ávalos, I., Mora-Medina, P., Olmos-Hernández, A., Verduzco-Mendoza, A., Reyes-Sotelo, B., & Martínez-Burnes, J. (2020). Advances in infrared thermography: Surgical aspects, vascular changes, and pain monitoring in veterinary medicine. *Journal of Thermal Biology*, 102664. <https://doi.org/10.1016/j.jtherbio.2020.102664>
- Cilulko, J., Janiszewski, P., Bogdaszewski, M., & Szczygielska, E. (2013). Infrared thermal imaging in studies of wild animals. *European Journal of Wildlife Research*, 59(1), 17–23. <https://doi.org/10.1007/s10344-012-0688-1>
- Cook, N. J., Smykott, A. B., Holm, D. E., Fasenko, G., & Church, J. S. (2006). Assessing feather cover of laying hens by infrared thermography. *The Journal of Applied Poultry Research*, 15(2), 274–279. <https://doi.org/10.1093/japr/15.2.274>
- De Boer, P. T., Kroese, D. P., Mannor, S., & Rubinstein, R. Y. (2005). A tutorial on the cross-entropy method. *Annals of Operations Research*, 134, 19–67. <https://doi.org/10.1007/s10479-005-5724-z>
- Decina, C., Berke, O., van Staaveren, N., Baes, C. F., & Harlander-Matuschek, A. (2019). Development of a scoring system to assess feather damage in Canadian laying hen flocks. *Animals*, 9(7), 436. <https://doi.org/10.3390/ani9070436>
- Giersberg, M., Spindler, B., & Kemper, N. (2017). Assessment of plumage and integument condition in dual-purpose breeds and conventional layers. *Animals*, 7(12), 97. <https://doi.org/10.3390/ani7120097>
- Glatz, P. C. (2001). Effect of poor feather cover on feed intake and production of aged laying hens. *Asian-Australasian Journal of Animal Sciences*, 14(4), 553–558. <https://doi.org/10.5713/ajas.2001.553>
- He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision* (pp. 2961–2969). <https://doi.org/10.1109/ICCV.2017.322>

- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778). <https://doi.org/10.1109/CVPR.2016.90>
- Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7132–7141). <https://doi.org/10.1109/CVPR.2018.00745>
- Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4700–4708). <https://doi.org/10.1109/CVPR.2017.243>
- Huang, E., Mao, A., Liu, K., & Xue, Y. (2020). Deep learning-based assessment of laying-hen feather conditions using color and thermal images. In *2nd asian conference on precision livestock farming (ACPLF2020)*. <http://acplf2020.csp.escience.cn/dct/page/1>
- Joulin, A., Cissé, M., Grangier, D., & Jégou, H. (2017). Efficient softmax approximation for GPUs. In *International conference on machine learning* (pp. 1302–1310). <https://doi.org/10.48550/arXiv.1609.04309>
- Lamping, C., Derks, M., Koerkamp, P. G., & Kootstra, G. (2022). ChickenNet—an end-to-end approach for plumage condition assessment of laying hens in commercial farms using computer vision. *Computers and Electronics in Agriculture*, 194, Article 106695. <https://doi.org/10.1016/j.compag.2022.106695>
- Lesson, S., Leeson, S., & Walsh, T. (2004). Feathering in commercial poultry II. Factors influencing feather growth and feather loss. *World's Poultry Science Journal*, 60(1), 52–63. <https://doi.org/10.1079/WPS20034>
- Ma, N., Zhang, X., Zheng, H. T., & Sun, J. (2018). Shufflenet v2: Practical guidelines for efficient cnn architecture design. In *Proceedings of the European conference on computer vision* (pp. 116–131). https://doi.org/10.1007/978-3-030-01264-9_8
- Main, D. C., Mullan, S., Atkinson, C., Bond, A., Cooper, M., Fraser, A., & Browne, W. J. (2012). Welfare outcomes assessment in laying hen farm assurance schemes. *Animal Welfare*, 21(3), 389–396. <https://doi.org/10.7120/09627286.21.3.389>
- McCafferty, D. J. (2013). Applications of thermal imaging in avian science. *Ibis*, 155(1), 4–15. <https://doi.org/10.1111/ibi.12010>
- Mota-Rojas, D., Titto, C. G., de Mira Geraldo, A., Martínez-Burnes, J., Gómez, J., Hernández-Ávalos, I., Casas, A., Dominguez, A., José, N., Bertoni, A., Reyes, B., & Pereira, A. M. (2021). Efficacy and function of feathers, hair, and glabrous skin in the thermoregulation strategies of domestic animals. *Animals*, 11(12), 3472. <https://doi.org/10.3390/ani11123472>
- Nääs, I. de A., Romanini, C. E. B., Neves, D. P., Nascimento, G. R. do, & Vercellino, R. do A. (2010). Broiler surface temperature distribution of 42 day old chickens. *Scientia Agricola*, 67(5), 497–502. <https://doi.org/10.1590/s0103-90162010000500001>
- Pichová, K., Bilčík, B., & Košťál, L. (2017). Assessment of the effect of housing on feather damage in laying hens using IR thermography. *Animal*, 11(4), 661–669. <https://doi.org/10.1017/s1751731116001981>
- Raine, B., Martin, J., Mitchell, M., Futro, A., & D'eath, R. (2020). Infrared thermography provides an accurate assessment of feather condition in broiler chickens. In *WPSA/BSAS UK 2020 spring meeting*. <https://doi.org/10.1080/17466202.2020.1917210>
- Redaelli, V., Ludwig, N., Costa, L. N., Crosta, L., Riva, J., & Luzi, F. (2014). Potential application of thermography (IRT) in animal production and for animal welfare. A case report of working dogs. *Annali dell'Istituto Superiore di Sanità*, 50, 147–152. <https://doi.org/10.4415/ANN.14.02.07>
- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4510–4520). <https://doi.org/10.1109/CVPR.2018.00474>
- Savory, C. J. (1995). Feather pecking and cannibalism. *World's Poultry Science Journal*, 51(2), 215–219. <https://doi.org/10.1079/WPS19950016>
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision* (pp. 618–626). <https://doi.org/10.1109/ICCV.2017.74>
- Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 1–14. <https://arxiv.org/abs/1409.1556>
- Spindler, B., Weseloh, T., Eßer, C., Freytag, S. K., Klambeck, L., Kemper, N., & Andersson, R. (2020). The effects of UV-A light provided in addition to standard lighting on plumage condition in laying hens. *Animals*, 10(6), 1106. <https://doi.org/10.3390/ani10061106>
- Tauson, R., Ambrosen, T., & Elwinger, K. (1984). Evaluation of procedures for scoring the integument of laying hens—Independent scoring of plumage condition. *Acta Agriculturae Scandinavica*, 34(3), 400–408. <https://doi.org/10.1080/00015128409435409>
- Wang, C. Y., Bochkovskiy, A., & Liao, H. Y. M. (2023). YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7464–7475). <https://doi.org/10.1109/CVPR52729.2023.00721>
- Xu, D., Shu, G., Liu, Y., Qin, P., Zheng, Y., Tian, Y., Zhao, X., & Du, X. (2022). Farm environmental enrichments improve the welfare of layer chicks and pullets: A comprehensive review. *Animals*, 12(19), 2610. <https://doi.org/10.3390/ani12192610>
- Zhang, X., Zhang, Y., Geng, J., Pan, J., Huang, X., & Rao, X. (2022). Feather damage monitoring system using rgb-depth-thermal model for chickens. *Animals*, 13(1), 126. <https://doi.org/10.3390/ani13010126>
- Zhao, Y., Xin, H., & Dong, B. (2013). Use of infrared thermography to assess laying-hen feather coverage. *Poultry Science*, 92(2), 295–302. <https://doi.org/10.3382/ps.2012-02629>